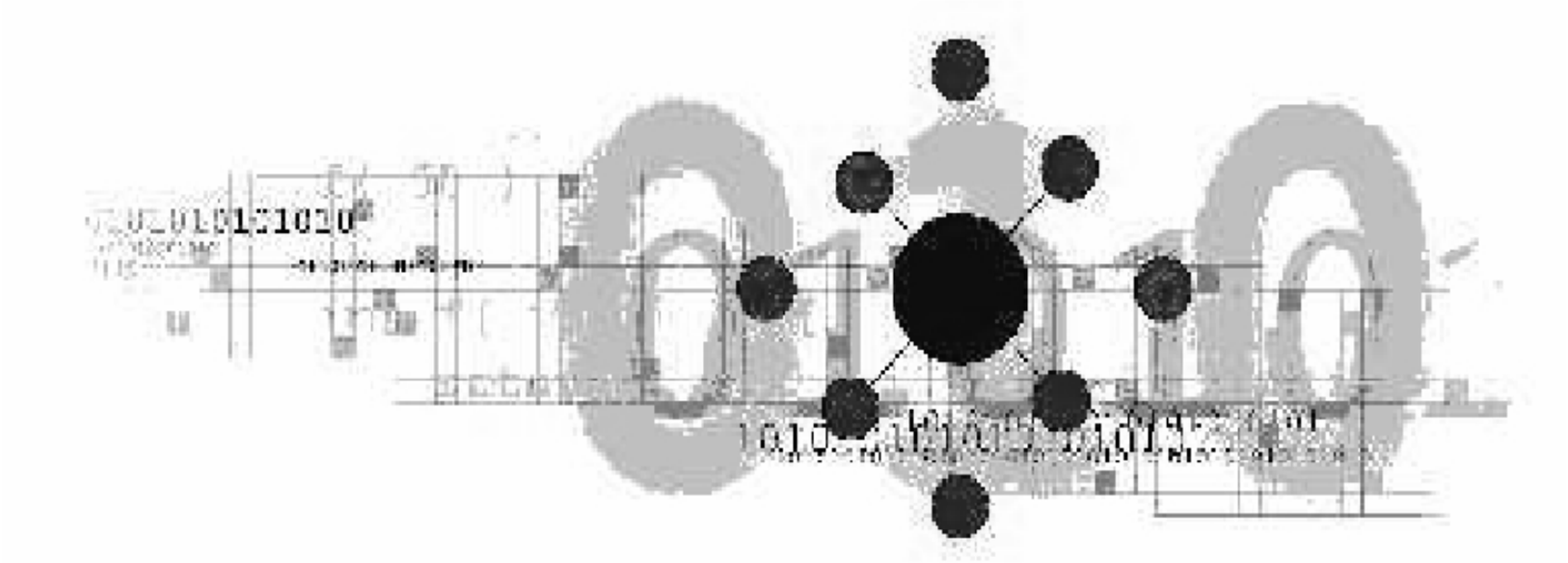
IRIX® 6.5.15 Update Guide



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Dear Valued Customer,

SGI is pleased to present the new IRIX 6.5.15 maintenance and feature release. Starting with IRIX 6.5, SGI created a new software upgrade release strategy, which delivers both the maintenance (6.5.15m) and feature (6.5.15f) streams. This upgrade is part of a family of releases that enhance IRIX 6.5.

There are several benefits to this strategy: it provides periodic fixes to IRIX, it assists in managing upgrades, and it supports all platforms. Additional information on this strategy and how it affects you is included in the updated *Installation Instructions* manual contained in this package.

If you need assistance, please visit the Supportfolio Online Web site at `http://support.sgi.com` or contact your local support provider.

In conjunction with the release of IRIX 6.5.10, SGI adopted expanded life cycle management categories to customize the services we deliver to our users. We now offer seven (7) modes of service on SGI software: Active, Maintenance, Legacy, Retired, Courtesy, Divested, and Expired.

Active Mode is our highest level of service and applies to products that are being actively developed and maintained. Software fixes for all levels of problems can be expected. Active Mode is for the current release. Prior releases are in Maintenance Mode or Legacy Mode.

Maintenance Mode software is maintained and is still an important part of our product mix, but no new functionality is added to it. Functional fixes for severe problems are generally available. Software stays in this category about 18 months before moving to the next service mode, Legacy.

Legacy Mode products are still supported through regular software support contracts. This software generally runs on out-of-production platforms, and is generally available and renewable. Software support functions are limited to providing existing fixes and "work arounds" for new problems. IRIX 6.2, IRIX 6.3 and IRIX 6.4 are good examples of software in this category.

Retired Mode allows us to provide restricted support for some older software products. This level of support has severe limitations on portability to new platforms, and availability of support at this level is limited to existing customers with whom we have existing contractual obligations. It is the final stage before the product is removed from the SGI product set. For example, IRIX 5.3, Gauntlet Firewall, and SoftWindows95 are all currently maintained at this level.

Courtesy Mode products are not officially part of the software support offerings, and call center support is not available. There may be a customer communications e-mail alias with varying questions and response times. It can be withdrawn at any time. IRIS Showcase and Cosmo Worlds are all examples of software in the Courtesy Mode of service.

Divested Mode - CA Unicenter TNG and Syntax TAS are examples of divested products, that is, products that have been turned over to a third party who assumes all responsibility for support. Calls received at the SGI Call Center for support on divested products will be redirected to the third party.

Expired Mode - Products like IRIXPro Proconf and Molecular Inventor, which have reached the end of their useful life, are placed in Expired Mode. These products are not supported or distributed in any form by SGI.

The Web page at
`http://support.sgi.com/irix/news/index.html#swmodes`
contains a link to a Web page describing in more detail all seven software support modes and links to the latest customer letters containing a list of the various software products that have changed support modes.

We thank you for your continued commitment to SGI.

Terry Oberdank
Vice President
Global Field Customer Services
SGI

Welcome to your SGI IRIX 6.5.15 update. This booklet contains:

- A list of key features in IRIX 6.5.15
- A list of CDs contained in the IRIX 6.5.15 update kit
- A guide to SGI Web sites

IRIX 6.5.15 Key New Features

The following changes have been incorporated into the core IRIX 6.5.15 overlay CDs and the Applications CD.

New Hardware Platforms or Features

- **SGI Origin 300 server series**
  Support for the SGI Origin 300 server series is included in 6.5.15. This includes support for the Origin 300 Base Module in 2, 4, and 8 processor sizes, the addition of a NUMAlink Module for scaling up to 32 processors, and the TP900 Storage Module.

New Software Features - Feature Stream Only

Filesystems

- **XVM Subvolume naming convention**
  For the IRIX 6.5.15f release, the default naming convention for subvolumes in the `/dev/lxvm/` and `/dev/cxvm` directories is *volname*, *subvolume*.

  Older releases of XVM created a directory entry for a subvolume of the form *volname_subvolname*. This convention can yield potential problem. For example, since `vol1_data` is a legal name for a volume it is impossible to determine whether `/dev/lxvm/vol1_data` refers to the data subvolume of the volume `vol1` or to a volume

named `vol1_data`. The *volname_subvolname* form of subvolume directory entries is still supported in IRIX 6.5.15f, but its use is not recommended.

For information on XVM device directories and pathnames, see the *XVM Volume Manager Administrator's Guide*.

- **Group quotas for XFS filesystems**
  For the 6.5.15f release of IRIX, you can implement disk quotas on XFS filesystems according to group ID. Previous releases of IRIX supported implementing quotas according to user and project ID. Project and group accounting are mutually exclusive. This feature is on-disk compatible with Linux-XFS group accounting, where this feature is already active. For information on administering XFS quotas, see *IRIX Admin: Disks and Filesystems*.

  Caution: Group quotas are supported in the feature stream only. If you implement group quotas on a disk and, subsequently, mount that disk with the `pquota mount` option on a machine running the maintenance stream or an earlier release of the feature stream on which group quotas are not supported, the quota accounting could be corrupted.

- **CXFS new features**
  IRIX 6.5.15f provides the following new features for CXFS.

- **Support for clients of other operating systems**
  Support has been added for clients of other operating systems such as Solaris as defined in *CXFS Client Administration Guide*. These clients are released asynchronously from the IRIX release. This support requires IRIX 6.5.15f plus appropriate patches. For more information, contact your SGI support contact.
- **NFS export of CXFS filesystems**
  Default scripts are now provided in the `/var/cluster/clconfd-scripts` directory to permit NFS-exporting of CXFS filesystems listed in `/etc/exports`.
- **rotatelogs script syntax**
  Changes have been made to the `rotatelogs` script syntax. The root `crontab` file now has an entry to run the `rotatelogs` script weekly. If you run the script twice in one day, it will append the current log file

- **CXFS upgrade: filesystem definitions**
  The structure of the CXFS filesystem configuration was changed with the release of IRIX 6.5.13f. Backward compatibility with earlier versions is no longer maintained as of IRIX 6.5.14f, since all nodes in the cluster must be running the same or adjacent releases.
  - If you are upgrading from 6.5.13f to 6.5.14f or later, there is no further impact.
  - If you are upgrading from 6.5.12f or earlier, you must perform a one-time manual conversion of your CXFS filesystem definitions.

See the *CXFS Software Installation and Administration Guide* for more information.

- **CXFS upgrade: cluster nodes**
  For CXFS customers, if you install IRIX 6.5.15f including `sysadm_base` on a cluster node where the 6.5.13f or earlier version of CXFS is installed, you will need to upgrade to the 6.5.15f version CXFS (`sysadm_cxfs`) and install 6.5.15f `sysadm_cluster`.

New Software Enhancements - Maintenance and Feature Stream

Filesystems

- **inode allocation**
  XFS inode numbers are 64-bit values containing an encoded disk location. For large filesystems (1 Terabyte plus), the inode numbers can overflow into the top 32 bits of the number. Certain backup applications which use the inode number only allow 32 bits of space, and thus have severe problems dealing with very large XFS filesystems.

  In IRIX 6.5.15, XFS has been changed to allocate inodes only within the lower portion of the filesystem to avoid this overflow issue. Other allocation policies in the filesystem will make this lower order space preferred for inodes and reduce the risk of a filesystem refusing to create new inodes when it still has space available.

For existing large filesystems with high numbered inodes this does not solve the problem, but it does solve it for new filesystems, and for new files created on existing filesystems.

For backwards compatibility a new mount option is available, inode64, this will allow xfs to place inodes anywhere in the filesystem and follow exactly the original placement policies. Filesystems which were small enough not to be able to overflow 32 bits of inode number also retain the old behavior.

Disk Support

- **Disk capacity**
  The 6.5.15 release of IRIX adds support for disks that have a capacity of 2 terabytes. Previously, the maximum was 1 terabyte. The maximum disk capacity had become an issue with the introduction of 180GB drives, multiples of which can be combined into a single logical unit (lun) behind a RAID controller.

  The maximum capacity of a logical unit for both of our current RAID controllers is 2 terabytes, so this change allows us to support that maximum.

System Configuration

- **Documentation of chproj(1)**
  For the IRIX 6.5.15 release, Chapter 5 “System Administration in a Multiuser Environment” in *IRIX Admin: System Configuration and Operation* has been updated with a new section on Project ID

numbers and the `chproj`(1) command that changes the project ownership of a file. The project ID may be either a project name found in the project file, or a decimal project ID. For more information, see the `chown`(1) man page and the *IRIX Admin: System Configuration and Operation* manual.

- **Default value of gang_sched_off**
  In IRIX 6.5.15, the systune parameter `gang_sched_off` is turned on by default. This means that gangs will not be running on the system unless the system administrator sets `gang_sched_off` to 0. This differs from previous IRIX releases, for which gangs ran on the system automatically.

  This change has been made per field request because we have discovered that having gang scheduling on by default greatly diminishes system performance. Most customers have been running with it off.

- **coremask parameter**
  The IRIX 6.5.15 release adds a coremask parameter set in the `/var/sysgen/mtune/kernel` file to determine what kind of permissions a core file will receive. Currently, when a process creates a core file, it uses the umask of the owner of the process to determine the permissions. This can lead to core files containing sensitive information being created with world readable permissions. This systune variable allows the system administrator to limit access to core files. The default is 0, which is the current behavior. However, if

the coremask variable is set to some other value, it uses that number as the umask and ignores the umask of the owner of the process. For example, if coremask is set to 0177 and the owner of the process has a umask of 022, the permissions set on the core file will be 600 instead of 644. This variable assumes that the number entered is octal. The default setting is 0. The minimum setting is 0. The maximum setting is 0177.

For more information on the coremask parameter, see Appendix A, "IRIX Kernel Tunable Parameters", in *IRIX Admin: System Configuration and Operation*.

- **New partmgr(1M) command**
  A new partitioning administration command, `partmgr`(1M), is available to define and store partition definitions. For more information, see *IRIX Admin: System Configuration and Operation*.

Networking

- **/dev/poll device added**
  The IRIX 6.5.15 release adds support for the `/dev/poll` interface for scalable and efficient event notification. For information on `/dev/poll`, see the `poll`(7M) man page.

  In order to use the new `/dev/poll` device in IRIX 6.5.15, you must run `/dev/MAKEDEV poll` out of `/dev` after you have rebooted your system with the 6.5.15 kernel. You need to do this only once, to set up the interface to the driver.

- **DHCP Setup**
  The "Dynamic Host Configuration With Proclaim" section in *IRIX Admin: Networking and Mail* has been updated to include detailed procedures and descriptions for setting up DHCP.
- **New DHCP client options**
  For the IRIX 6.5.15 release, the DHCP client (`proclaim`) has been updated to include options to allow DHCP clients to capture changes to DNS, NIS, or GATEWAY in your environment. For more information, see the `proclaim`(1M) man page and *IRIX Admin: Networking and Mail*.

System Availability

- **IRIX 6.5.15 and FailSafe 2.1.2**
  For FailSafe customers, if you install IRIX 6.5.15, including `sysadm_base`, on a cluster node where FailSafe 2.1.1 is installed, you will need to upgrade to FailSafe 2.1.2 (`sysadm_failsafe2`) and install 6.5.15 `sysadm_cluster`.

Scalability

- **Designating CPUs in the cpuset configuration file**
  The IRIX 6.5.15 release provides the ability to designate one or more CPUs or a range of CPUs on a single line in the `cpuset` configuration file. The CPUs in a `cpuset` configuration file do not have to be specified in a particular order.

The following is a sample configuration file that describes an exclusive `cpuset` containing 7 CPUs:

```
# cpuset configuration file
EXCLUSIVE
MEMORY_LOCAL
MEMORY_EXCLUSIVE

CPU 16
CPU 17-19, 21
CPU 27
CPU 25
```

For more information on the `cpuset` configuration file, see Chapter 4, “Cpuset System”, in *IRIX Admin: Resource Administration*.

System Performance and Tuning

- **Update to systune(1M) man page**
  For the 6.5.15 release, the `systune`(1M) man page has been updated to describe a situation where the system segment size value (`syssegsz`) reported by the `systune` command is different than the value set in the `/unix` file as follows:

  The `systune`(1M) may not show the user-defined value of `syssegsz` after a system reboot. This is due to the fact that the colormap must be calculated and added to the `syssegsz` value for machine types IP19, IP20, IP22, and IP32. When `systune` is called, it compares the value of `syssegsz` in the kernel to the value set in the `/var/sysgen/stune` file. If this is done on one of the above

machines, the values will not match because `syssegsz` has been altered with the colormap value in the kernel, but not in the `/var/sysgen/stune` file. This indicates normal functioning of the operating system.

Origin 300 Systems

- **Support for PCI Expansion Module on Origin 300 systems**
  The IRIX 6.5.15 release introduces support for the PCI Expansion Module to Origin 300 systems. The PCI Expansion Module allows PCI expansion on an Origin 300 independent of the addition of additional CPUs and memory. Each PCI Expansion Module adds 12 PCI slots and up to 4 PCI Expansion Modules to the system overall.

SGI Origin 3000 server (Onyx 3000 server)

- **Partitioned FLEXlm licensing change**
  Starting with the IRIX 6.5.15 release, each partition of a partitioned Origin 3000 system will have a new unique FLEXlm host ID (`lmhostid`). Any FLEXlm license based on the old non-partitioned `lmhostid` value will no longer work and a new license key must be generated. See the following web page for additional details: `http://www.sgi.com/support/licensing/`.
  - Contact the applicable software product vendor and provide the `lmhostid` number of the partition to generate a new license key. For nodelock licenses, provide the `lmhostid` for all partitions of the software installed and operating.

  - For software licensed by SGI, see the following web page for more details: `http://www.sgi.com/support/licensing/`.
  - For email and telephone contact information, see the IRIX 6.5.15 Welcome page and follow the "Bundled software and licenses" link.

- **HOTPLUG support for attach and detach**
  For IRIX 6.5.15 we have added HOTPLUG support for attach and detach. This feature is supported on IP35 Origin 3000 series systems only; it is not supported on IP35 0300 systems. For information on this feature, see *IRIX Admin: Networking and Mail*.

Licensing software

- **lmparthostids and lmbasehostid commands**
  The IRIX 6.5.15 release supports the `lmparthostids` and `lmbasehostid` commands. The `lmparthostids` command generates a list of all 63 partition specific `hostids` and the unpartitioned id. This list can be passed to a license key generation tool for nodelock licenses so that the license works across all partitions or an unpartitioned system. The `lmbasehostid` command takes any arbitrary `lmhostid` (this could be a partition specific value) and generates the base unpartitioned id. That number could be used when the base `lmhostid` is required, even from a partition.

- **Updated FLEXlm v7.2i support from GLOBEtrotter Software, Inc.**
  The license manager daemon and the floating license server are built in V5 compatibility mode. For backwards compatibility, the client libraries and header files are still based on FLEXlm v6.1g. For more information, see the *FLEXlm End User Manual* and the associated man pages.

SGI software product life cycle management

- **IRIX support policy**
  In accordance with the SGI software product life cycle management strategy announced in November 2000, SGI will announce support mode changes for various software products on a quarterly basis. To view customer letters containing support mode changes announced since November 2000, as well as the latest IRIX Support Policy documents, visit the IRIX Support Policy Web page located at:

  `http://support.sgi.com/news/support/index.html`

Applications CD

The following products were updated with bug fixes only:

- Netscape Communicator 4.79

  By default, Netscape Communicator version 4.79 will install the new N32 version of the browser. Any existing third-party plugins for the browser that are O32 will no longer function. Support for the O32 Macromedia Flash plugin is also included but requires the O32

Netscape client which must be manually selected for installation. For more information on the features of the Netscape client, see `http://browsers.netscape.com/browsers/main.tmpl`.

- IRIX Interactive Desktop System Administration
- Information Searching Execution Environment
- Impressario 2.10.5
- NEdit 5.1.1c
- Webviewer 3.0.2

The following products contain enhancements or new features:

- InSight Online Doc Viewer 4.4.1 is now N32 for better performance.
- SGI Web Server, based on the Apache 1.3.22 release

IRIX OS Bundled Software

- **SCSL 1.4.0.1**
  SGI Scientific Library 1.4.0.1 (SCSL 1.4.0.1) is now included on the IRIX 6.5.15 Overlays CDs. SCSL is a collection of high-performance mathematical and numerical libraries that are widely used in scientific and technical compute intensive applications.

  For more information on SCSL, see `http://www.sgi.com/software/scsl.html`.

For more information about the bundled software that is included with this release, see CD Contents and the Bundled Software and Licenses web page that you can access from the Welcome web page.

Freeware CD

The following changes have been incorporated for the IRIX 6.5.15 release time frame.

New products for this release:

- cdrtools 1.10 - audio/data CD recording tools
- elisp-manual-21-2.7 -Emacs-21 Lisp reference manual
- host_991529 - DNS query tool with rfc1876 support
- fftw-2.1.3 - Library for computing the Discrete Fourier Transform
- jabber-1.4.1 - An Instant Messaging Server
- netcdf-3.5.0 - Unidata Network Common Data Form (netCDF) library
- pdflib-4.0.1 - Library for generating PDF files
- python-2.1.1 - Python Language Interpreter
- sono-1995aug11 - Postscript sonogram generator (replaces fw_NotamSono)
- t1lib-1.3 - Type 1 font rasterizer
- tnef-1.1.1 - Unpack application/ms-tnef archives

- xpdf-0.93 - PDF file viewer

Updated products for this release:

- ImageMagick-5.4.1-2 - X11 image display/manipulation
- ORBit-0.5.12 - CORBA Object Request Broker
- amanda-2.4.2p2 - Network backup system
- analog-5.03 - Web Server Log File Analyser
- bison-1.30 - GNU bison (improved yacc)
- bonobo-1.0.17 - GNOME compound document library
- bzip2-1.0.1 - Compress/decompress files
- cvs-1.11.1p1 - Concurrent Versions System
- elm2.5.6 - curses based mail agent
- emacs-21.1 - GNU extensible text editor
- gcombust-0.1.47 - GUI for cdrtools/cdlabelgen
- gdb-5.1 - GNU debugger
- gimp-1.2.2 GNU Image Manipulation Program
- gnuplot-3.7.1 - Plot graphs from data
- grace-5.1.4 - 2D plotting
- groff-1.17.2 - Document Formatting System

- gtkglarea-1.2.3 - OpenGL widget for GTK+
- htmldoc-1.8.16 - HTML indexer/converter
- imap-2001a - UW imap and pop daemons
- indent-2.2.7 - GNU C code formatter
- libpcap.0.6.2 - User level packet capture
- libungif-4.1.0b1 - Uncompressed gif image utils
- mozilla.0.9.6 - Mozilla for IRIX
- mozilla-src.0.9.6 - Mozilla source code for IRIX
- mysql-3.23.44 - MySQL relational database
- nas-1.4.2 - network audio system
- oaf-0.6.7 - GNOME Object Activation Framework
- openssh-3.0.2p1 - Secure rsh, rcp replacements
- Perl - Extensible scripting language (v5.6.1)
- php-4.0.6 PHP - Hypertext Processor http-server scripting
- proftpd-1.2.4 - FTP Daemon
- sablot-0.71 - Sablotron XML/XSLT library, release 0.71
- samba - Samba Version 2.2.2
- screen-3.9.10 - GNU virtual terminal manager
- socks5-v1.0r11 - socks5 proxy server and clients

- sudo-1.6.3p7 - Restricted configurable ‘su’
- transfig.3.2.3d - xfig image format converters
- wget-1.7.1 - WWW site mirroring tool
- wu-ftpd-2.6.2 - WU FTP Daemon
- xdemineur-2.1.1 - X11 based minesweeper game
- xfig.3.2.3d - X11 based drawing program
- xmorph-2001jul27 - X11/GIMP image morphing program
- xpat2-1.07 - X11 solitaire card games
- xplanet-0.90 - X11 background image generator
- xscreensaver-3.34 - Screen Saver and Locker
- xtraceroute-0.9.0 - OpenGL based traceroute GUI

Products that were omitted for this release:

- bladeenc-0.91 - generate MP3 files from WAV or AIFF sound files (now part of fw_cdrtools)
- cdrecord-1.8 - record audio or data CDs
- elisp-manual-20-2.5 - Emacs-20 Lisp reference manual
- mkhybrid-1.12b5.3 - ISO9660/JOLIET/HFS/RockRidge systools (now part of fw_cdrtools)
- mkisofs-1.12 - iso9660 filesystem tools (now part of fw_cdrtools)

- mozilla-src.0.9.4 - Mozilla source code for IRIX (available on Freeware website, `http://freeware.sgi.com/`)
- Notam's Sono audio score maker (fw_NotamSono replaced by fw_sono)

For more information about the freeware that is included with this release, including a complete list of the freeware packages, see `http://freeware.sgi.com/`.

For more information about the bundled software that is included with this release, see CD Contents and the Bundled Software and Licenses Web page that you can access from the Welcome Web page.

Documentation

The following manuals have been revised for the significant new features incorporated into this release.

IRIX operating system

- 007-2825-010, *IRIX Admin: Disks and Filesystems*, was updated to include information on imposing disk quotas on XFS filesystems according to group IDs.
- 007-2860-007, *IRIX Admin: Networking and Mail*, includes an updated and expanded section on "Dynamic Host Configuration with Proclaim."
- 007-3700-010, *IRIX Admin: Resource Administration*, was updated to include information about the `cpuset` configuration file.
- 007-2859-018, *IRIX Admin: System Configuration and Operation*, was updated to include information on the `coremap` kernel parameter, the `partmgr` command, and using the `chproj` command to change the project ID of a file.
- 007-3723-009, *Upgrading an IRIX Operating System on a /target_root*, was updated with some clarifications to the procedures it provides.

Filesystems and Volume Managers

- 007-4016-012, *CXFS Software Installation and Administration Guide*, was updated to include information on new features and procedures. See

the “New Features in This Guide” section of the manual for a full description of the changes.

- 007-4003-008, *XVM Volume Manager Administrator's Guide*, was updated to include information on the new XVM subvolume naming convention.

Additional manuals

- 007-4365-001, *Origin 300 User's Guide*, was made available to support the SGI Origin 300 server series.

Key New Features from IRIX 6.5.1 to IRIX 6.5.14

Hardware Platforms and Features Supported

No new hardware support introduced with IRIX 6.5.14

Introduced in IRIX 6.5.13:

- Support for the 500MHz R14K processor on SGI Origin 2000 and Onyx2 systems

Introduced in IRIX 6.5.12:

- Support for the 500MHz R14K processor on SGI Origin 3000 series and Onyx 3000 systems

No new hardware support introduced with IRIX 6.5.10 and 6.5.11

Introduced in IRIX 6.5.9:

- Support for the SGI 3000 series of servers, including the SGI 3200, SGI 3400, and SGI 3800 servers
- Support for the TVO digital video option board for Silicon Graphics Onyx2 systems
- Support for the QED RM7000 processor on O2 systems

Introduced in IRIX 6.5.8:

- Support initiated for VPro Graphics, the next generation graphics for Silicon Graphics Octane systems Introduced in IRIX 6.5.7:
- Support for Silicon Graphics Onyx2 InfiniteReality3 systems
- R12000S CPU on SGI 2200, SGI 2400, SGI 2800, SGI 2100, and Origin 200

Introduced in IRIX 6.5.5:

- QLA2200 (copper and optical) is supported for FC-AL, FC-AL via the Emulex hub, or fabric attach via the Brocade Silkworm 2000 switches

Introduced in IRIX 6.5.4:

- 270-MHz processor for Silicon Graphics O2 and Silicon Graphics Octane visual workstations
- HDTV XIO Board for Silicon Graphics Onyx2 and SGI Origin 2000 systems (this now includes the former Cray Origin 2000 system)

Introduced in IRIX 6.5.3:

- Systems using the MIPS R12000 processor
- Digital Video Multiplexer option board (DPLEX) for Silicon Graphics Onyx2 systems
- Redundant Power Supply (RPS)
- 21" Monitor Support for O2 and Octane systems

Introduced in IRIX 6.5.2:

- Flat Panel Monitor for O2 systems
- 16-pipe Onyx2 InfiniteReality systems
- GSN network adapter

Introduced in IRIX 6.5.1:

- Gigabit Ethernet for Octane and SGI Origin systems
- 128p Metarouter for Origin 2000 systems (formerly known as Cray Origin 2000 systems)
- Dual Channel Display option for O2 systems
- Onyx2 InfiniteReality2 systems
- 225QC for SGI Origin 200 systems

New Software Features-Feature Stream Only

Introduced in IRIX 6.5.14

- The CXFS GUI was enhanced in the IRIX 6.5.14 release as follows:
  - **CXFS Manager** and **CXFS Cluster View** windows are now combined into one window called **CXFS Manager.**
  - Tasks can now be launched by clicking the right mouse button over the tree-view area, or by selecting **Tasks** on the menu bar.

  - Command line interfaces (CLIs) that the GUI runs can be viewed from a new **File > Show SALog** menu item.
  - You can now partially mount filesystems on just a subset of nodes, using the **Define Filesystem** and **Modify Filesystem** tasks.
  - A new **Find** text field helps you find items within the displayed tree-view area. Filesystem status and cluster status update faster.

  For more information, see the *CXFS Software Installation and Administration Guide*.

- The structure of the CXFS filesystem configuration was changed with the release of IRIX 6.5.13f. Backward compatibility with earlier versions is no longer maintained as of IRIX 6.5.14f, since all nodes in the cluster must be running the same or adjacent releases.
  - If you are upgrading from 6.5.13f to 6.5.14f or later, there is no further impact.
  - If you intend to run a mixture of 6.5.13f and 6.5.14f nodes, you must turn off backward compatibility.
  - If you are upgrading from 6.5.12f or earlier without first installing and running 6.5.13f, you must perform a one-time manual conversion of your CXFS filesystem definitions.

  See the *CXFS Software Installation and Administration Guide* for more information.

- For CXFS customers, if you install IRIX 6.5.14f including `sysadm_base` on a cluster node where the 6.5.13f version of CXFS is installed, you will need to upgrade to the 6.5.14f version CXFS (`sysadm_cxfs`) and install 6.5.14f sysadm_cluster.

Introduced in IRIX 6.5.13

- With 6.5.13, the structure of the CXFS filesystem configuration has been changed. CXFS filesystems can now be defined, modified, managed and deleted independently of each other, and of the cluster definition.

  (Previously, the CXFS filesystems were defined as attributes to the cluster definition.) To accommodate clusters mixing nodes running 6.5.12 and 6.5.13, backwards compatibility is enforced by default in 6.5.13.

  For more information, see the *CXFS Software Installation and Administration Guide.*

- The XVM Volume Manager can be used when layered with the CXFS filesystem. The XVM Volume Manager can also be used as a standalone volume manager; this requires that you be running the 6.5.13f release leg of the IRIX operating system. The 6.5.13m leg does not support XVM as a standalone volume manager; this support will be added in a later release. For information on CXFS filesystems, see *CXFS Software Installation and Administration Guide*.

- Added support for the use of mirrors in XVM logical volumes. The mirroring feature of XVM requires the XFS Volume Plexing software option. Customers running CXFS and who want to run mirrors will need to purchase this license. XLV customers with plexing licenses can upgrade to XVM without having to acquire a new license.

Introduced in IRIX 6.5.12

- Supports the labeling of disks as XVM system disks in the XVM Volume Manager. This lets you create XVM logical volumes that include the partitions of a system disk. The following XVM system disk features are supported:
  - Root partitions can be mirrored
  - There can be multiple root partitions on a system disk
  - You can include usr and swap partitions in any XVM logical volume configuration, including mirrors, concats, and stripes
  - A system disk can include slices that are not part of a root, usr, or swap partition

  You can now convert existing system disks to XVM system disks, and then use their partitions as part of an XVM logical volume. After you have converted an existing system disk to an XVM disk, you can convert the disk back to its original state by unlabeling the disk with the XVM unlabel command. For information on XVM system disks, see the *XVM Volume Manager Administrator's Guide*.

Introduced in IRIX 6.5.10:

- XVM installed with CXFS is now qualified on IP35 (Origin 3000) systems in addition to the existing qualification on the IP27 (Origin 200, Origin 2000, and Onyx2) and IP30 (Octane) systems.

Introduced in IRIX 6.5.9:

- CXFS supports the use of hierarchical storage management (HSM) products through the data management application programming interface (DMAPI). An example of such a product is SGI's Data Migration Facility (DMF).
- CXFS and IRIS FailSafe 2.1 can be installed and run on the same system, which is known as coexecution. This provides application-level high availability and a clustered filesystem with nodes that support CXFS services, FailSafe services, or both.

  Note: The CXFS and FailSafe coexecution feature is not available in the IRIX maintenance stream. IRIS FailSafe 2.1 is an independently shipped product and is not shipped with IRIX.
- Support for the Scheduled Transfer Protocol (STP). STP is a new lightweight network protocol that uses Direct Memory Access (DMA) to read or write data into user space from a network interface. This lets high bandwidth devices, such as Gigabyte System Network (GSN) and Gigabit Ethernet (GigE), perform at network speeds with minimum interrupt overhead. For more information, see the `stp`(7P) man page or `http://www.hippi.org`.

- Support for disk quotas that can now be set by project ID. Disk quotas let you limit the amount of space a user or project can occupy and the number of files (inodes) that each user or project can own. You can implement hard or soft limits; hard limits are enforced by the system and soft limits only remind the user to decrease disk usage.

  For more information on disk quotas and their administration, see *IRIX Admin: Disks and Filesystems*, which now includes information about project quota administration. For more information on project IDs, see *IRIX Admin: Backup, Security, and Accounting*. For more information on the administration of disk quotas by project, see the `edquota`(1M), `find`(1M), `quota`(1M), `repquota`(1M), and `fstab`(4) man pages.

- Support for the waitjob feature, which includes the new functions `setwaitjobpid`() and `waitjob`(). These functions let the batch schedulers query job information following job termination. When a batch scheduler launches a job, it calls `setwaitjobpid`() to tell the new job what pid is waiting for information upon termination. When the job terminates, it remains as a zombie until either the batch scheduler calls `waitjob`() to retrieve the job's termination information or the waiting pid no longer exists. The information returned includes the job start time, usage information, and reason for termination.

  For more information, see *IRIX Admin: Resource Administration* and the `waitjob`(1M) and `setwaitjobpid`(1M) man pages.

Introduced in IRIX 6.5.8:

- Support for Comprehensive System Accounting (CSA). CSA is a set of user and administrative C programs and shell scripts that provide methods to collect per-process resource usage data, monitor disk usage, and charge fees to specific login accounts. CSA uses this per-process accounting information and combines it by job identifier within the system boot uptime periods. CSA provides the following features that are not available with any other IRIX accounting package:
  - Per-job accounting
  - Daemon accounting (tape and Network Queuing System (NQS))
  - Flexible accounting periods (daily and periodic accounting reports can be generated as often as desired and not restricted to once per day or once per month)
  - Flexible system billing units (SBUs)
  - User exits for site specific customization of daily and periodic accounting
  - Configurable parameters within the `/etc/csa.conf` file
  - User job accounting (`ja` command)
  - For more information, see *IRIX Admin: Resource Administration* and the `csa`(1M) man page.

Introduced in IRIX 6.5.7:

- Support for job limits

  Job limits allow system administrators to manage user access to system resources by setting limits on different system usage parameters. This can result in improved system throughput and utilization. For more information on the usage of the different system usage parameters, see the *IRIX Admin: Resource Administration Guide* and the following man pages: `jstat`(1), `jlimit`(1), `job_limits`(5) `genlimits`(1M), and `showlimits`(1).

- Support for the CXFS product

  CXFS provides a cluster file system that allows file sharing between machines. CXFS includes the following capabilities: high resiliency and availability, reduced storage costs, and scalable high performance. The initial version of CXFS was introduced with IRIX 6.5.6.

  CXFS requires a new volume manager, XVM. XVM installed with CXFS is currently qualified only on IP27 (Origin 200, Origin 2000, and Onyx2) and IP30 (Octane) systems. The base cluster XVM software is packaged with IRIX 6.5.6f and above. Optional XVM features will be separately licensed and are targeted to release in the IRIX 6.5.10f and above time frame.

The IRIX 6.5.7 release version of CXFS contains metadata server recovery. See the *CXFS Software Installation and Administration Guide* for the administrative shutdown procedures and additional troubleshooting information.

See the "New for 6.5.7 (or above): Caveats before you Install" section at the Caveats to Read Before You Upgrade link at `http://support.sgi.com/6.5/start_here/doc657/precaveats.html` for known dependencies or limitations before installing CXFS with XVM. Also, the Caveat and Release Note Updates link at `http://support.sgi.com/6.5/caveat_updates.html` should be checked periodically for information on the latest CXFS and XVM patches, descriptions of implemented CXFS and XVM features, updated Release Notes, or any late-breaking caveats.

Introduced in IRIX 6.5.6:

- Support for the Scheduled Transfers (ST) protocol. The ST protocol is an ANSI standard level two through four protocol suite designed to support extremely high performance data movement. ST provides a socket-based interface to applications, which lets you directly port the existing networking applications to ST. ST also supports the OS Bypass mode of operation, which allows smaller messages to be sent and received with extremely low latencies. In this initial release of ST, only the Gigabyte System Network (GSN) network adapter is supported.

Introduced in IRIX 6.5.2:

- Motif 2.1/IRIS ViewKit 2.1
- IRIX Oplock support
- Support for non EUC encoding and locales (`sjis/big5/gbk/utf8`)

Maintenance and Feature Streams

Introduced in IRIX 6.5.4:

- In the IRIX 6.5.5 release, SGI introduced a new optional directory format for XFS filesystems, referred to as the "version 2" directory format in `mkfs` output.

  XFS filesystems with a version 2 directory format can provide improved reliability and performance over filesystems with a version 1 directory format in some applications, particularly for applications that use NFS exported XFS filesystems.

  The directory format for an XFS filesystem is specified with the -n parameter of the mkfs command. The default directory format is currently the original version 1 directory format.

  Beginning with the IRIX 6.5.14 release, version 2 directories of XFS will be the default for all new filesystems created with `mkfs`.

  Version 1 directory filesystem creation will still be supported, but this will require that you specify the `-n version=1` parameter of the `mkfs` command.

SGI recommends that all new XFS filesystems be created with version 2 directories. IRIX releases older than IRIX 6.5.5, however, will not be able to mount a filesystem created with a version 2 directory and will issue the following message when a mount is attempted:

Wrong filesystem type: xfs

Many SGI customers are currently running production servers with version 2 directories. If you want to format filesystems with the version 2 directory format prior to the 6.5.14 release, you can specify the following option of `mkfs`:

```
mkfs -n version=2
```

There is no means for converting a filesystem, in place, between version 1 and version 2 directories. A filesystem can be converted between version 1 and version 2 directories by means of an `xfsdump/mkfs/xfsrestore` sequence.

For more information, see *IRIX Admin: Disks and Filesystems*.

- A `-q` *cpuset_name* `-p` option was added to the `cpuset` command that allows you to see the properties of particular cpuset, such as, the number of processes and CPUs associated with the specified cpuset.
- The `cpusetGetProperties()` and `cpusetFreeProperties()` functions were added to the Cpuset System library. The `cpusetGetProperties()` function allows you to retrieve various properties associated with a cpuset. The `cpusetFreeProperties()` functions allows you to release the

memory used by a `cpuset_Properties_t` structure. Any programmer that uses the cpuset API from Irix 6.5.14, and later, can use these functions. For information, see chapter 4, "Cpuset System", in the *IRIX Admin: Resource Administration* manual and the `cpusetGetProperties`(3x) and `cpusetFreeProperties`(3x) man pages.

- Parts of the *IRIX Interactive Desktop* were refreshed in IRIX 6.5.14. While functionality has not changed, the navigation, imagery, and graphics design have been enhanced in the following areas:
  - The main IRIX login window (clogin) has a new design using new SGI colors, font, and logo. The product is called sysadmdesktop and appears automatically at boot when root has `chkconfig noiconlogin on`.
  - **EZsetup** has a new layout and login icon using new SGI colors, font, and logo. The product is called sysadmdesktop and is launched by logging into **EZsetup account** from main IRIX login window.
  - The **System Manager** main window uses new SGI colors, font, and logo. The product is sysadmdesktop and is launched from the **System toolchest**.
  - The Welcome to SGI web pages have new navigation and new SGI colors, font, and logo. The product is Welcome. Launch from **Find toolchest**, select **WhatsNew**, double-click the **Welcome_to_SGI** icon.

  - The Background Setting panel includes seven new desktop patterns using the new SGI logo and font: Camouflage, Citrus Citrus, Iron, Midnight, Reef, Sand, and Sprinkle. Note: You may notice slow performance when previewing all but the Sprinkle background, but not when switching desks (see the `background`(1) man page). Old backgrounds are still available. The product is desktop_eoe and is launched from the **Desktop toolchest**, select **Customize**, then **Backgrounds**.

Introduced in IRIX 6.5.13

- The `xfsrestore` command returns an incorrect exit code when it encounters a media error; improving this process lets a system administrator intelligently react to these conditions.
- Improved exit codes for the `xfsrestore` and `xfsdump` commands. This lets a system administrator correctly respond to end-of-media or operator interrupt conditions.
- Changed the `mkfs` command to allow you to specify the size of an XFS allocation group, as an alternative to specifying the total number of allocation groups. You use the `-d agsize= option` for this. For information, see the `mkfs_xfs` man page and *IRIX Admin: Disks and Filesystems*.
- Changed the `mkfs` command to allow you to specify the size of a stripe unit and the size of a stripe width in bytes or in filesystem blocks, as an alternative to specifying these values in 512-byte block

units. You use the `-d su=` and the `-d sw= options` for this. For information, see the `mkfs_xfs` man page and *IRIX Admin: Disks and Filesystems*.

- Changed the default size of an XFS allocation group; larger filesystems will result in larger default allocation group sizes.
- The `xfsdump` and `xfsrestore` commands will provide the VSN of the tape that reached its end-of-volume (or the VSN of a new tape that needs to be mounted) and pass this VSN to the `media_change_alert_program` specified with the `-c` option. This lets system administrators send commands to a tape jukebox to mount the next tape.
- Changed the default size of an XFS log. The default log size grows with the size of the filesystem up to the maximum log size, 128 megabytes, on a 1 terabyte filesystem. For information on the default values of XFS allocation groups and XFS log sizes, see *IRIX Admin: Disks and Filesystems*.
- Added `cpusetAttachPID()` and `cpusetDetachPID()` functions to the Cpuset System library. The `cpusetAttachPID()` function allows a programmer to attach a currently running program to an existing cpuset. The `cpusetDetachPID()` function allows a programmer to detach a currently running program to an existing cpuset. Any programmer that uses the cpuset API from Irix 6.5.13, and later, can use these functions. For information, see chapter 4, "Cpuset System", in the *IRIX Admin: Resource Administration* manual

and the `cpusetAttachPID`(3x)and `cpusetDetachPID`(3x) man pages.

- The following features have been moved from Enhanced DHCP to standard DHCP:
  - Ping check before an address is leased to insure it is not already in use
  - MAC address filtering; clients whose MAC address is listed in a file will be unable to obtain a lease and/or accesses from these clients will be logged.

  For more information, see the `dhcp_bootp` man page.
- Added implementation of End Sequence to Scheduled Transfer Protocol. The upper layer protocol or applications can now terminate a transfer and restart another one without taking down virtual connection.
- VPro for Octane/OpenGL performance improvements have been added as follows:
  - Improved X window performance by 25 percent based on `x11perf`
  - Improved performance of OpenGL `glCopypixel()` function
- The following new system controller features have been implemented for the SGI Origin 3000 server series:

  - Enabled router port security, which prevents unauthorized enabling of additional router ports on router bricks in the SGI Origin/Onyx 3400 server series
  - Enabled system serial number security, which prevents unauthorized changes to the system serial number
  - Completely redesigned the graphical interface for the front-panel display of the SGI 3400 and 3800 servers

- Added hotplug attach feature that allows Gigabit Ethernet cards to be installed on a running system. A system administrator can use the `pciconfig` command to add cards to an Origin 3000 system and the `ifconfig` command to configure the cards for networking. For more information, see the `pciconfig` and `ifconfig` man pages.
- Added Origin 3000 system partitioning support in the `eoe.sw.partition` software package. This feature improves how system partitioning software is installed, enabled, and disabled. It also eliminates the need to manually edit configuration files.
- Efficiency improvements have been made to the BTE driver layer to reduce both latency and contention. This feature is a performance improvement to the existing functionality that is used internally by the system.
- The tools that report CPU utilization have been modified to correct the sampling of %WIO to omit idle time in line with industry standards. See the `sar(1)` man page for details.

- Updated FLEXlm v7.2f support from GLOBEtrotter Software, Inc. The license manager daemon and the floating license server are built in V5 compatibility mode. For backwards compatibility, the client libraries and header files are still based on FLEXlm v6.1g. For more information, see the *FLEXlm End User Manual* and the associated man pages.

Introduced in IRIX 6.5.12

- OpenGL Performer 2.4.1 execution environment, which is a powerful suite of tools and features that creates real-time visual simulation and other professional performance-oriented 3D graphics applications for the IRIX operating system. The following new features for OpenGL Performer provide enhanced realism and peak-performance rendering:
  - Programmable model shading using pfShader
  - Use of pfVolFog for rendering layered and patchy fog
  - Use of pfRotorWash to create the visual effect of downwash in helicopter simulation
  - Double-precision matrix support for rendering extremely large databases where objects are far away from the origin
  - CPU priority specification to allow multiple run-time processes to run on the same processor
  - Multipipe scalability enhancements

  - Improved pfLOD behavior to allow custom evaluation functions
  - Speed and functionality enhancements for pfFlux, pfCliptexture, and pfLightpoints
  - pfHyperpipe support for the DPLEX hardware option
  - EventView performance tuning tool for instrumenting and tuning your OpenGL Performer application
- The IRIX tape-support feature consists of a tape support driver, personality daemons, and a daemon that manages the personality daemons. The feature enables new personality daemons to be developed and added individually. The tape-support driver, which coexists with the current IRIX tape driver (tpsc), passes all device-dependent tasks, except for I/O processing, to a personality daemon. This release supports various devices including the Fibre Channel implementation on the StorageTek 9840 and T9940A tape drives. The `reserve`, `release`, `preempt`, `clear`, and `prsv` commands are available for these tape drives on the `mt` command. For more information, see the `mt`(1), `tsarchive`(1M), `tsdaemon`(1M), `tserrpt`(1M), `tsset`(1M), `tsstop`(1M), `ts.config`(4), and `ts`(7) man pages.
- Updated FLEXlm v7.2e support from GLOBEtrotter Software, Inc. The license manager daemon and the floating license server are built in V5 compatibility mode. For backwards compatibility, the client libraries and header files are still based on FLEXlm v6.1. For more

information, see the *FLEXlm End User Manual* and the associated man pages.

- The Scheduled Transfer Protocol (STP) has moved from the feature stream to the maintenance and feature stream. STP is now a mature supported protocol and it includes TRIX support.

Introduced in IRIX 6.5.11:

- Support for the use of a reserved area of memory; this reserved area is between virtual addresses 0x30000000 and 0x40000000. This feature disables the reserved area of memory that the kernel provides for applications to place their mmap and shmat mappings. For more information, see the `mmap`(2), `shmat`(2), `syssgi`(2), and `sgi_use_anyaddr`(1) man pages.
- Support for weightless pthreads; they allow you to lower the priority of a process by using the `npri -w` command. For more information, see the `npri`(1) man page.
- Support for upgrading license_eoe to the FLEXlm 7.1 release from Globetrotter Software, Inc. The license manager daemon will be built in V5 compatibility mode. For more information, see the *FLEXlm End User Manual* and the associated man pages.
- Support for an increased maximum number of logical unit numbers (LUNs) in the SCSI infrastructure from 64 to 255. This change impacts only the fcadp and qlfc fibre channel drivers and they will now enforce this LUN limit. If a LUN is greater than 255, it will be ignored

after issuing a message to this effect. If you use large numbers of LUNs, you can exhaust the hardware graph. You may need to use the `systune` command to increase the hwgraph_num_dev parameter to let the kernel build larger hardware graphs.

- Support for licensing of partitioned environments. The SGI Origin 3000 server series and SGI Onyx 3000 series of systems give you the flexibility to distribute and manage a partitioned system environment on a large CPU server system. In most cases, software licensing behaves in the same manner on a single system image as in a partitioned environment. However, there are some differences. Depending on your applications, you may need either systemwide or per-CPU licensing. For more information, see *IRIX Admin: Software Installation and Licensing*.
- Support for the PCI Hot Plug infrastructure on the SGI 3200, SGI 3400, and SGI 3800 servers. The Hot Plug feature consists of the Hot Insertion and Hot Removal actions. Each device driver must be qualified as supporting one or both of these actions. Currently, the QLogic fibre channel and QLogic SCSI device drivers are qualified for Hot Insertion. For more information, see the `pciconfig`(1) man page for specific administrative options.
- Support for HP LaserJet 4050 and HP LaserJet 8100 printers
- OpenGL Performer Runtime 2.2.11 overlay incorporates the latest fixes. The Performer 2.2.11 overlay can be installed from the

`/CDROM/dist/` directory of the IRIX 6.5.11 Overlays CD (3 of 3), February 2001.

Introduced in IRIX 6.5.10:

- Support for point-to-point connections between the QLogic 2200 fibre channel controller and the Brocade switch. This type of connection lets the QLogic 2200 controller perform full duplex transfers with the switch offering a potential increase in bandwidth. This capability was enabled by changing the value assigned to the *qlfc_use_connection_mode* variable (in `/var/sysgen/master.d/qlfc`) to 2. The previous value was 0. This change does not impact loop operation.
- Support for the IRIX SCSI tape driver (TPSC) enhancements that let system administrators and privileged applications specify a persistent reservation on shared tape drives. After the reservation has been established, the shared tape drive cannot be accessed by another host until the reservation has been released. For more information, see the `mt`(1) and `mtio`(7M) man pages.
- Support for the new Fortran, C++, and base compiler runtime libraries packaged under the `ftn_eoe`, `c++_eoe` and `compiler_eoe` overlay images. These libraries are based on the MIPSpro 7.3.1.2m compilers and contain new interfaces released under the MIPSpro 7.3 compilers. The libraries are backward compatible with the libraries that were released in previous versions

of IRIX 6.5. For more information, see the `ftn_eoe`, `c++_eoe`, and `compiler_eoe` release notes.

- OpenGL Performer Runtime 2.2.10 overlay incorporates the latest fixes. The Performer 2.2.10 overlay can be installed from the `/CDROM/dist/` directory of the IRIX 6.5.10 Overlays CD (3 of 3), November 2000.

Introduced in IRIX 6.5.9:

- Support for Embedded Support Partner 2.0 (ESP2.0). This release supercedes patch 3895 and includes these additional major features as follows: Support for Embedded Support Partner 2.0 (ESP2.0). This release supercedes patch 3895 and includes these additional major features as follows:
  - Fixes to the System Group Manager for secure interconnectivity between group members and the group manager
  - Support for an unlimited number of group members to the System Group Manager
  - Multiple ESP user support with page privileges
  - ESP logbook for electronically logging and tracking support activities locally on the system

  The migration from ESP1.0 to ESP2.0 is done seamlessly when upgrading from previous versions of IRIX to IRIX 6.5.10.

The Embedded Support Partner Overview and the Embedded Support Partner User Guide have been combined. The new document is called the Embedded Support Partner User Guide and will contain overview and usage information for the ESP 2.0 command line and graphical user interfaces.

- Support for the SCSI Enclosure Services (SES). A new subsystem that manages the Fibre Channel drive enclosures has been added. The new daemon, `sesdaemon`, supports the Xyratex 12-slot enclosure and the Clarion 10-slot enclosure. The new client application, `sesmgr`, uses a command set similar to the set used by the previous `fcagent/fccli` subsystem. For more information, see the `sesdaemon`(1M) and `sesmgr`(1M) man pages.
- OpenGL Performer Runtime 2.2.9 overlay incorporates the latest fixes. The Performer 2.2.9 overlay can be installed from the `/CDROM/dist/` directory of the IRIX 6.5.9 Overlays CD (3 of 3), Support for the SGI Origin 300 server series is included in 6.5.14. This includes support for the Origin 300 Base Module in 2, 4, and 8 processor sizes, the addition of a NUMAlink Module for scaling up to 32 processors, and the TP900 Storage Module. 2000.

Introduced in IRIX 6.5.8:

- Embedded Support Partner (ESP) patch 3895 migrates ESP 1.0 to ESP 2.0. This patch release includes new features and bug fixes. The major features are as follows:

  - Automatic Call Logging to the SGI call centers for Mission Critical Supported customers
  - Loading of specific event profiles for monitoring
  - A new command line interface to setup and use ESP
  - A new user interface with a high level of usability and navigational features
  - Fixes to the System Group Manager

  The Embedded Support Partner Overview and the Embedded Support Partner User Guide will be combined. The new document is called the Embedded Support Partner User Guide and will contain overview and usage information for the ESP 2.0 command line and graphical user interfaces.

- Support for 32-bit direct mapping to any node on the system. This feature lets the system administrator change the 32-bit direct mapping node for a specific Peripheral Component Interconnect (PCI) bus. It also provides a new interface, `pcibr_get_dmatrans_node()`, that lets a device driver obtain the node ID. For more information, see the *IRIX Device Driver Programmer's Guide* and the `pcibr_get_dmatrans_node`(D3) man page.
- Support for the Cpuset System programming interface application (previously known as `miser_cpuset`). This interface is provided in the form of a Dynamic Shared Object (DSO). You can use this

programming interface to create cpusets, remove cpusets, and attach processes to cpusets. You can use the Cpuset System programming interface in areas where the `cpuset` command is inappropriate. For example, if a batch system needs to use the cpuset capability in IRIX, the Cpuset System programming interface will provide a more flexible and robust solution than the `cpuset` command. For more information, see *IRIX Admin: Resource Administration* and the `cpuset`(5), `cpusetAllocQueueDef`(3x), `cpusetAttach`(3x), `cpusetCreate`(3x), `cpusetDestroy`(3x), `cpusetDetachAll`(3x), `cpusetFreeCPUList`(3x), `cpusetFreeNameList`(3x), `cpusetFreePIDList`(3x), `cpusetGetCPUCount`(3x), `cpusetGetCPUList`(3x), `cpusetGetName(`3x), `cpusetGetNameList`(3x), and `cpusetGetPIDList`(3x) man pages.

- Support for new Miser cpuset options. These options allow the creation of restrictive memory pools from the nodes that contain the CPUs listed in the configuration file.

  Processes that exceed the available memory on those nodes may be terminated or paged (selectable). For more information on these options, see the `cpuset`(4) man page.

- OpenGL Performer Runtime 2.2.8 overlay incorporates the latest fixes. The Performer 2.2.8 overlay can be installed from the `/CDROM/dist` directory of the IRIX 6.5.8 Overlays CD (3 of 3), May 2000.

- Documenter's Workbench with the latest Y2000 bug fixes can be installed from the `/CDROM/dist/unbundled` directory of the IRIX 6.5.8 (or above) Overlays CD (2 of 3), May 2000.

Introduced in IRIX 6.5.7:

- Updating `sendmail` to version 8.9.3

  The version of the IRIX sendmail mail system supplied on this release and previous IRIX 6.5.*x* releases is based on `sendmail` version 8.8.8. Due to customer demand, the current sendmail.org release, `sendmail` version 8.9.3 (see `http://www.sendmail.org`), will be supplied via patch 3865 or its successor. Check the Caveat and Release Note Updates link at `http://support.sgi.com/6.5/caveat_updates.html` for the IRIX 6.5.7 late breaking caveats to determine the availability of this patch on Supportfolio. SGI intends to support `sendmail` 8.9.3 (or above) as the standard released `sendmail` in a future IRIX release, target 6.5.10.

  There are many differences between IRIX `sendmail` version 8.8.8 and version 8.9.3.

  – The major difference is their configuration files. The configuration file in `sendmail` version 8.9.3 is configured with the `sendmail.mc` file which is processed using the m4 macro processor to create the `sendmail.cf` file.

  - A new version of `configmail` configures the `sendmail.mc` file and provides features similar to the `configmail` utility in previous versions of IRIX. This version of `configmail` also processes the `sendmail.mc` file into `sendmail.cf` by using the m4 macro processor.
  - One of the new features included in version 8.9.3 and in great demand by IRIX users is the anti-relay features which can be used to control spam messages.

  For more information on the 8.9.3 version of `sendmail`, see the *IRIX Administration: Networking and Mail Guide* provided with the patch. For more information on how to configure `sendmail` 8.9.3, see `http://www.sendmail.org/m4/readme.html`.

- Open Inventor Runtime 2.1.6 overlay incorporates fixes included in versions 2.1.4 and 2.1.5. The Inventor 2.1.6 overlay can be installed from the `/CDROM/dist/unbundled` directory of the IRIX 6.5.7 Overlay CD 2 of 2. This overlay requires that the base Inventor 2.1.4 image be installed already, or with the overlay during the same install session. The Inventor 2.1.6 overlay can be installed on IRIX 6.5.5 and later.
- OpenGL Performer Runtime 2.2.7 overlay incorporates the latest fixes. The Performer 2.2.7 overlay can be installed from the `/CDROM/dist/` directory of the IRIX 6.5.7 Overlays CD (2 of 2), February 2000.

Introduced in IRIX 6.5.6:

- Support for a multithreaded version of the automatic filesystem mount daemon `autofsd`. This enhanced functionality allows for simultaneous multiple automounts. If a particular server for an automounted filesystem is not running or is slow to respond, one `autofsd` thread can wait for that server while other `autofsd` threads mount filesystems from other servers. This capability improves the automount performance and simultaneously provides longer wait times for downed servers, which should lead to a decrease in automount failures.

Introduced in IRIX 6.5.5:

- Embedded Support Partner, which is an integral part of the IRIX operating system, provides system administrators with a way to monitor various events (such as system events, changes in system hardware and software configuration, and system performance) on their systems. Embedded Support Partner is a set of daemons that perform the monitoring activities. These include an event monitoring daemon (`eventmond`), an event management daemon (`espemd`), and a database daemon (`espdbd`). Embedded Support Partner provides single-system monitoring capabilities as a standard part of IRIX. Optionally, Embedded Support Partner can be configured to receive event and system configuration data from all systems contained within a system group. Embedded Support Partner is controlled through a Web browser that is connected to the

Configurable Web Server, which is included in the Embedded Support Partner package. For more information, see the *Embedded Support Partner Overview*, the *Embedded Support Partner User Guide*, and the Embedded Support Partner man pages.

- Support for the version 2 XFS directory format; this format lets you choose a filesystem block size to match the distribution of data file sizes without adversely affecting directory operation performance. The directory format is specified with the `-n` parameter of the `mkfs` command. For more information, see *IRIX Admin: Disks and Filesystems* and the `mkfs_xfs`(1M) man page.
- Support for the math and scientific library SCSL 1.3. SCSL 1.3 will replace `Challengecomplib` on all supported system platforms at the time of the next major IRIX Release.

  SCSL provides support for the math and scientific libraries and is widely used in scientific and technical compute-intensive applications. SCSL 1.3 incorporates all the current `Challengecomplib` 3.1 features and will be distributed as a separately packaged product until the next major IRIX release.

  SCSL 1.3 can be downloaded from the SGI Download Cool Software Web page at `http://www.sgi.com/Products/Evaluation`. If you do not have Web access and are a current support customer, you can request CD media free of charge through your local support center. Non-contract customers can contact their sales representatives to order SCSL 1.3.

New features introduced in addition to the `Challengecomplib` functionality since the release of SCSL 1.1 are:

- Added convolution/correlation and filter routines to the signal-processing functionality (formerly available only in `Challengecomplib`)
- Improved ordering techniques for the sparse linear solvers
- Performance enhancements for the MIPS R12000 processor
- Bug fixes from SCSL 1.1 and 1.2

Challengecomplib entered maintenance mode with the release of IRIX 6.5.5. No new features or enhancements will be incorporated.

For more information on SCSL, see `http://www.sgi.com/software/scsl.html`

- Support for Automated Performance Monitoring. Together with Embedded Support Partner, the base performance monitoring services in the pcp_eoe product have been extended to include an inference engine for evaluating rules about system-level performance and raising alarms. Also provided is a parameterized set of standard rules that can be selectively enabled and tuned to meet local requirements and to choose alternative alarm notification mechanisms. These features are of most value to operations staff running production IRIX systems. For more information, see the `pmie`(1) and `pmieconf`(1) man pages, and the *Performance Co-Pilot IRIX Base Software Administrator's Guide*.

- Two new options were added to the `miser_create_cpuset` command. These options allow additional restrictions on memory assignment for processes running on a CPU set. These options are documented in the `miser_cpuset`(4) man pages.

Introduced in IRIX 6.5.4:

- Support for the Miser queue repack policy. When a job finishes execution before the end of its schedule, the system resources it was using are released. This policy attempts to reschedule the jobs using earlier start and end times to take advantage of these released system resources. The order of the scheduled jobs will be maintained. This feature can be used by all Miser users running IRIX 6.5.4m or f and later releases. For more information, see the `miser`(4) and `miser`(5) man pages, and *IRIX Admin: System Configuration and Operation*, Chapter 7 "Managing User Processes."
- Distributed Computing Environment (DCE) Client for accessing shared resources in distributed computing DCE/DFS serving environments
  - Kernel libraries only
  - Requires installation of DCE/DFS 1.2.2a software for full functionality

Introduced in IRIX 6.5.3:

- (Octane systems only) The worst-case interrupt response time is guaranteed to be less than one millisecond on properly configured Octane systems
- Support for the X security and appgroup extensions (combined with a new Netscape plug-in, these allow the embedding of X applications in Web pages)
- Support for European fonts, including the Euro currency symbol
- New Software Manager and Inst commands to simplify selections for upgrades
- New Software Manager and Inst configuration variable to more easily handle cases where configuration files are upgraded. See the *smart_config_handling* preference in `inst` or `swmgr` for more information.
- Support for LDAP 3.0

Introduced in IRIX 6.5.2:

- AutoFS extended to use UNS for map information
- Support for DCShare application sharing extension
- Fibre channel support to Dmnet
- Three new HP printer drivers: HP4000, HP5000 and HP4500 (Color LaserJet 4500DN)

Documentation changes

Manuals in the IRIX Admin document set are updated when necessary to document new features at each IRIX release. The front of each manual includes a description of new features and major documentation changes for the current revision, as well as a record of when the manual was revised. The most recent version of an IRIX Admin document available on the Technical Publications Library includes information for features available in the current IRIX release.

The Techpubs Library is available at: `http://techpubs.sgi.com`.

Applications CD changes

Note: Applications take two forms: full images and overlays. The base versions of each can be found on the Applications CD. When full images are updated, new versions are placed on the Applications CD. Upgrades of overlay products, however, are located on the Overlay.

Introduced with IRIX 6.5.14

The following products were updated with bug fixes only:

- Bug fix update for SGImeeting 2.0.1
- Final bug fix update for Color Management
- Bug fix update for Internet Gateway 3.2
- Security and other bug fixes for NEdit 5.1.1b

- Minor fixes for Appletalk 10.01
- Netscape Communicator 4.78

  By default, version 4.78 will install the new N32 version of the browser. Any existing third-party plugins for the browser that are O32 will no longer function. Support for the Macromedia Flash plugin is also included. For more information, see `http://browsers.netscape.com/browsers/main.tmpl`.

Introduced with IRIX 6.5.13

The following products were updated with bug fixes only:

- SGI Web Server, based on the Apache 1.3.20
- IRIX Interactive Desktop System Administration
- Information Searching Execution Environment
- InSight Online Doc Viewer 4.4
- Netscape Communicator 4.77

  By default, version 4.77 will install the new N32 version of the browser. Any existing third-party plugins for the browser that are O32 will no longer function. Support for the Macromedia Flash plugin is also included. For more information, see `http://home.netscape.com/communicator/v4.5/tour/index.html`.

The following products contain enhancements or new features:

- WebSetup 3.3 has been updated to support the SGI web server based on Apache 1.3.20 and an administration GUI has been added.
- Impressario 2.9.5 to supports a new HP6300 Scanner driver and an Epson PPD bug fix.

Introduced with IRIX 6.5.12

- SGI web server, based on the Apache web server version 1.3.17, replaces the Netscape FastTrack web server; for details on how to change your default web server and configure and administer the SGI web server, see the Caveats to Read After You Upgrade area in the Start Here document for more details.
- WebSetup 3.2 has been updated to support the SGI web server based on Apache
- NEdit version 5.1.1a has added extensions to the regular expression syntax, an improved international character set input, new macro subroutines, and improved syntax highlighting for several programming languages
- Impressario 2.8.5 provides support for two new HP printer drivers, HP4050 and HP8100
- Runtime Plug-in for IRIX, Java Edition 1.1.1b has added support for the Netscape N32 plugin and the O32 plugin

- The following products from the Isogon Corporation have been removed: LicensePower/iFOR IS4.0.1 ARK and LicensePower/iFOR IS4.0.1 CRK. Sales and support has been divested to the Isogon Corporation at `http://www.isogon.com`. For more information on support modes, see the Support Policy page at `http://www.support.sgi.com/news/support/index.html#support_policy`.

Introduced with IRIX 6.5.11

- Cosmo Player, IRIX Interactive Desktop, Impressario, IRIX Interactive Desktop Tools, IRIX Interactive Desktop Administration, IRIS InSight Viewer, IRIS InSight Developer, and IRIS InSight Dynaweb Server, Netscape Communicator, Netscape Developer, and IRIS Showcase have been updated to incorporate bug fixes.
- AccessX has been updated with the latest revision of the help book. It can be viewed after installation of the product.
- Graphics Demonstration Programs, Graphics Demonstration Programs for O2, Octane Demos, and Octane Demos 2 software have been removed from the IRIX Apps CD and incorporated into existing platform specific Demonstration Program CDs that ship with new systems.
- Nedit GUI style editor has been updated from version 4.0.3 to the latest freeware version, 5.1.1.
- Netware 1.1.1 is now compatible with SGI 3000 systems.

- Webviewer has been revised to prepend the new threaded Cosmo Player plugin directory to the default webviewer plugin search path.

Introduced with IRIX 6.5.10:

- IRIX Interactive Desktop, IRIX Interactive Desktop Tools, IRIX Interactive Desktop Administration, IRIS InSight Viewer, IRIS InSight Developer, and IRIS InSight Dynaweb Server have been updated with bug fixes.
- Cosmo Player 2.1.4 release has added support for the Netscape N32 plugin.
- CustomerLink Client Software will no longer be supported. Its key features have been migrated to the SupportFolio Online site (`http://support.sgi.com`).
- Customer Support Services Base Software will no longer be supported. Its key features have been migrated to the SupportFolio Online site (`http://support.sgi.com`).
- Impressario 2.6.5 has added 1000 new PPD files to the `/usr/spool/lp/PPD_untested/` directory. The README file in this directory contains details on how to use the new drivers. The new PPD files include printers from the following manufacturers:

3M, Adobe, Agfa, Apple, Autologic, Canon, Dataproducts, Epson, FujiPhotoFilm, Hitachi Koki Co., Ltd., Hewlett-Packard, IBM, Kodak, Konica, Linotype, MGI, Mitsubishi, OKI, Optronics, PrePress, QMS, Ricoh, Samsung, Scitex, Shinko, Sony, Splash, Topmax, Tektronix, Xante, Xerox

- Netscape Communicator 4.75; by default, version 4.75 will install the new N32 version of the browser. Any existing third-party plugins for the browser that are O32 will no longer function. Support for the Macromedia Flash plugin is also included. For more information, see `http://home.netscape.com/communicator/v4.5/tour/index.html`. Netscape Developer and Netscape Lite have also been updated to version 4.75.
- Netscape FastTrack Server 3.03 contains an updated version of the Administration Server 3.52 (initially added in 6.5.5)
- Demonstration Programs, Octane2 demos added
- Runtime Plug-in for IRIX, Java Edition 1.1.1a has added support for the Netscape N32 plugin
- SGImeeting Collaboration Environment with Extensions, 2.0, Net-based data-conferencing, application-sharing software. A 30-day evaluation license is included. SGImeeting 2.0 is an update release to SGImeeting 1.4. Key new features includes:

- Support for Microsoft NetMeeting 3.0. Enhanced NetMeeting 3.0 interoperability - enables offline selection of compatibility for sharing applications supported by NetMeeting 3.0. For more information, see the SGImeeting Help and User's Guide.
- Improved application sharing performance - up to 80% from SGIMeeting 1.4 depending on your hardware configuration and internet connection. 80% improvement was achieved with 128MB, 180Mhz, R5000 O2 host talking to a 128MB 175Mhz R10000 Octane client over 100baseT ethernet sharing a maximized 1280x1024 solidview demo window.
- Speed dialing - enables creation of `.cnf` files and initiates calls from the command line.
- Background operation - provides selections for SGImeeting to minimize, remain actively connected in the background, and pop up when you receive a call.
- Grouped window sharing - enables selecting all windows with a specific X class name to be treated as a single application and shared as a group.
- Includes the SGIMeeting extensions (first included with SGIMeeting 1.4). The SGImeeting Extensions offer additional whiteboard tools as "plugins" that appear on the whiteboard tool palette. These tools are ideal for group discussions in CAD and image intensive industries. The following extensions are available:

- A screen capture tool for dynamic or still whiteboard input

- A video capture tool for dynamic or still whiteboard input

- Customizable arrows and dimension markers for easy discussion pointing

- Symbol palette for drag and drop images

- SmartClear for clearing annotations when captured images update

For more information on SGImeeting, see `http://www.sgi.com/software/sgimeeting`. For additional assistance, contact your local SGI sales representative.

– A time-limited demo is included with IRIX 6.5.10, contact your local SGI sales representative to purchase permanent licenses.

- Xinet Macintosh Connectivity Software, also referred to as Xinet Appletalk, now includes K-AShare, K-FS, and K-Spool software. These products provide enhancements for Macintosh file sharing, file serving, and printing connectivity with IRIX systems. These Xinet products were updated to version 10.02 in the IRIX 6.5.9 release. Version 10.02 is only the demonstration version; no license is included. Any licenses from previous versions of Xinet software will not work with version 10.02. For more information on Xinet products, see `http://www.xinet.com`. For technical or sales

questions, please contact Xinet at `sales@xinet.com` or 1.510.845.0555.

Introduced with IRIX 6.5.9:

- AccessX, Acrobat, Appletalk, Cosmo Player, IRIX Interactive Desktop System Administration, Impressario InSight, and Java have been updated with bug fixes.
- SGIMeeting 1.4 has been added back to the Apps CD and is provided with a 30 day Evaluation License that allows a demo collaborative session to run for 10 minutes.

  Here are some key changes for the 1.4 release:

  – The single `sgimeeting` image incorporates the prior base (`sgimeeting`) and extensions (`sgimeeting_ext`) functionality
  – Bug fixed since the 1.1 release are included
  – Purchased as an unbundled software product
  – Licensing is now required: available as a single license or a five license volume pack

  For more information on SGImeeting, see `http://www.sgi.com/software/sgimeeting`. For additional assistance, contact your local SGI sales representative.

Introduced with IRIX 6.5.8:

- No major feature or enhancements for this release

- AccessX, Desktop Runtime, Impressario, Infosearch, InSight, License Runtime, Sysadmin Desktop updated with bug fixes

Introduced with IRIX 6.5.7:

- SGImeeting and SGImeeting Extensions are now both licensed products and are no longer available on the Applications CD. Future revisions of the base SGImeeting product and the Extensions will be distributed on separate CDs and licensed separately. For additional information or assistance, contact your local SGI sales representative.
- Netscape 4.7A provides additional localization and the Flash plug-in since the 4.7 release.

Introduced with IRIX 6.5.6:

- Upgrade to Netscape Communicator 4.7; for details, see `http://home.netscape.com/communicator/v4.5`.

Note: Netscape Radio is implemented only for systems with the G2 player installed.

Introduced with IRIX 6.5.5:

- WebViewer Library Execution Only Environment 3.0

IRIX OS Bundled Software

Changes with IRIX 6.5.14:

- SCSL 1.4

  SCSL is a collection of high-performance mathematical and numerical libraries that are widely used in scientific and technical compute intensive applications. An iterative solver for sparse systems of linear equations, as well a direct solver for sparse complex symmetric systems of linear equations, were added to the SCSL 1.4 release. Additional features added to SCSL for this release are as follows:

  - A thread-safe parallel random number generator
  - Fortran90 interfaces to the BLAS
  - A parallel implementation of the solve phase for the out-of-core sparse solver
  - Removal of all entry points to the `CHALLENGEcomplib FFT` routines
  - Performance enhancements and rollup bug fixes from SCSL 1.3.

  SCSL can also be downloaded free of charge from the Download Cool Software page at `http://www.sgi.com/products/evaluation/`.

  For more information on SCSL, see `http://www.sgi.com/software/scsl.html`

For customers who do not have web access and are enrolled under a valid support contract, CD media can be requested free of charge from their local support center. Non-contract customers can order SCSL from their local sales representative.

Bundled Applications (ASE/AWE)

Changes with IRIX 6.5.10:

- EnlightenDSM is no longer included with IRIX 6.5 ASE and AWE

  SGI will no longer bundle EnlightenDSM with IRIX 6.5 ASE or AWE. The same software is now downloadable for free for all SGI customers from the Enlighten Software Solutions Web site, `www.enlightendsm.com/freedsm.html`. The version of EnlightenDSM bundled with IRIX 6.5 was the basic functionality. Please contact Enlighten with any product questions `www.enlightendsm.com`.

Changes with IRIX 6.5.8:

- Syntax TAS and CA Unicenter TNG Framework no longer included with IRIX 6.5 ASE

  SGI no longer bundles the Syntax TAS software with Origin systems as of June 2000. For interoperability solutions with PCs, Macintosh, and other systems, customers can contact Syntax at `www.syntax.com` or evaluate some of the solutions from SGI such

as Samba (SC4-SAMBA-2.0.7). For Macintosh interoperability, SGI offers a demo version of Xinet's KAShare (Appletalk) product (`www.xinet.com`) on the IRIX Applications CD.

- SGI no longer includes CA Unicenter TNG Framework with IRIX ASE. A free CD of CA Unicenter TNG Framework is now available from `www.ca.com/fw_reg.htm`.

  For more information about the bundled software that is included with this release, see CD Contents and the Bundled Software and Licenses web page that you can access from the Welcome web page.

IRIX 6.5.15 Update Kit Contents

The IRIX 6.5.15 Update Kit contains the following items for both server and workstation system configurations:

- CD Name:
    1. IRIX 6.5.15 (1 of 4) Installation Tools & Overlays CD, February 2002
    2. IRIX 6.5.15 (2 of 4) Installation Tools & Overlays CD, February 2002
    3. IRIX 6.5.15 (3 of 4) Overlays CD, February 2002
    4. IRIX 6.5.15 (3 of 4) Overlays CD, February 2002
    5. IRIX Applications for 6.5.15, February 2002
- The IRIX CD booklet *Installation Instructions: Installing an Intermediate (Overlay) Release, Installing Applications, Installing Software Licenses*

SGI Web Sites

IRIX 6.5-Related Web Sites

- SGI product information

  `http://www.sgi.com/products`

- IRIX 6.5 datasheet

  `http://www.sgi.com/software/irix6.5/datasheet.pdf`

- Start Here: Installing IRIX 6.5.14

  `http://support.sgi.com/6.5/installing.html`

- To view all qualified applications compatible with IRIX 6.5 releases, see `http://support.sgi.com/6.5/spk`

Services

- Customer Education Services

  `http://www.sgi.com/support/custeducation/`

- Professional Services

  `http://www.sgi.com/services`

Online Tools

- Customer Registration

  `http://www.sgi.com/support/custreg.html`
- Software Licensing/Key-O-Matic

  `http://www.sgi.com/Support/Licensing`
- Supportfolio Online

  `http://support.sgi.com`
- Online documentation - Technical Publications Library

  `http://techpubs.sgi.com`

Other Sites

- Download Cool Software

  `http://www.sgi.com/Products/Evaluation`
- Free Software

  `http://freeware.sgi.com`

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